PROBLEM SET 1

2. Linear Equations and the Output Gap

Orlando Almazán Gallardo

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The IGAE (in spanish: Indicador Global de la Actividad Económica) is a monthly indicator that measures the overall level of economic activity in Mexico. In simple words, it reflects how active the economy is at a given point in time. This indicator is commonly assumed to admit a decomposition into two components: a trend and a cyclical component,

$$y_t = \tau_t + c_t, \tag{1}$$

where y_t is typically defined as the logarithm of the observed data, that is,

$$y_t = \log(\mathrm{IGAE}_t). \tag{2}$$

The trend component τ_t is estimated using a regression model of the form

$$\tau_t = \beta_0 T_0(x_t) + \beta_1 T_1(x_t) + \cdots, \tag{3}$$

where $T_k(\cdot)$ are functions that ensure that the estimated trend is a smooth curve.

The output gap is then defined as the difference between the observed series and the estimated trend,

$$\widehat{\mathrm{gap}}_t = y_t - \widehat{\tau}_t. \tag{4}$$

Equivalently, if one writes the model as

$$y_t = \tau_t + \varepsilon_t, \tag{5}$$

where ε_t captures the cyclical fluctuations around the trend, the estimated output gap corresponds to the residuals of the regression:

$$\widehat{\mathrm{gap}}_t = y_t - \widehat{y}_t = e_t. \tag{6}$$

Thus, the output gap can be interpreted as a measure of how far the economy is from its trend at a given time. A positive gap indicates that economic activity is above trend, while a negative gap indicates that it is below trend.

The objective of this document is to model the output gap using Chebyshev polynomials (first of degree 7 and then of degree 51) and to compare the resulting estimates of the 7-th degree polynomial with the output gap estimates provided by Banco de México.

(1) Output gap for Mexico using a Chebyshev polynomial of order 7.

As mentioned in Bierens (1997, pp. 31–32), Chebyshev polynomials are well suited for modeling nonlinear functions such as trends. A Chebyshev polynomial of order k is defined as

$$P_{0,n}(t) = 1, \tag{7}$$

$$P_{k,n}(t) = \sqrt{2}\cos\left(\frac{k\pi(t-0.5)}{n}\right), \qquad k = 1, 2, \ldots \tag{8}$$

In the context of time series, t represents time, n is the sample size, and k denotes the degree of the polynomial. These polynomials are orthogonal, which makes them a useful alternative for mitigating multicollinearity problems in regression models.

Thus, when Chebyshev polynomials are used as regressors, the resulting model can be written as

$$y_1 = \beta_0 + \beta_1 P_{1,n}(1) + \cdots + \beta_m P_{m,n}(1) + \varepsilon_1, \tag{9}$$

$$y_2 = \beta_0 + \beta_1 P_{1,n}(2) + \cdots + \beta_m P_{m,n}(2) + \varepsilon_2, \tag{10}$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 P_{1,n}(n) + \cdots + \beta_m P_{m,n}(n) + \varepsilon_n, \tag{11}$$

where $y_t = \log(\text{IGAE}_t)$.

In MATLAB, using the Gauss–Jordan algorithm and solving the normal equations

$$(X'X)\boldsymbol{\beta} = X'y, \tag{12}$$

where X is the regression design matrix, the following coefficient estimates are obtained:

Coefficient	Estimate
β_0	4.4039
β_1	-0.1719
β_2	-0.0274
β_3	-0.0241
β_4	-0.0178
β_5	-0.0124
β_6	0.0033
β_7	-0.0142

Table 1: Estimated coefficients using a 7th- degree Chebyshev polynomial.

Using these estimates, the regression residuals are computed as

$$e_t = y_t - \widehat{y}_t, \tag{13}$$

which, as mentioned before, correspond to the estimated output gap.

Thus, by modeling the trend with a Chebyshev polynomial of degree 7, Figure 1 displays the estimated monthly output gap for Mexico from 1993 to 2025.

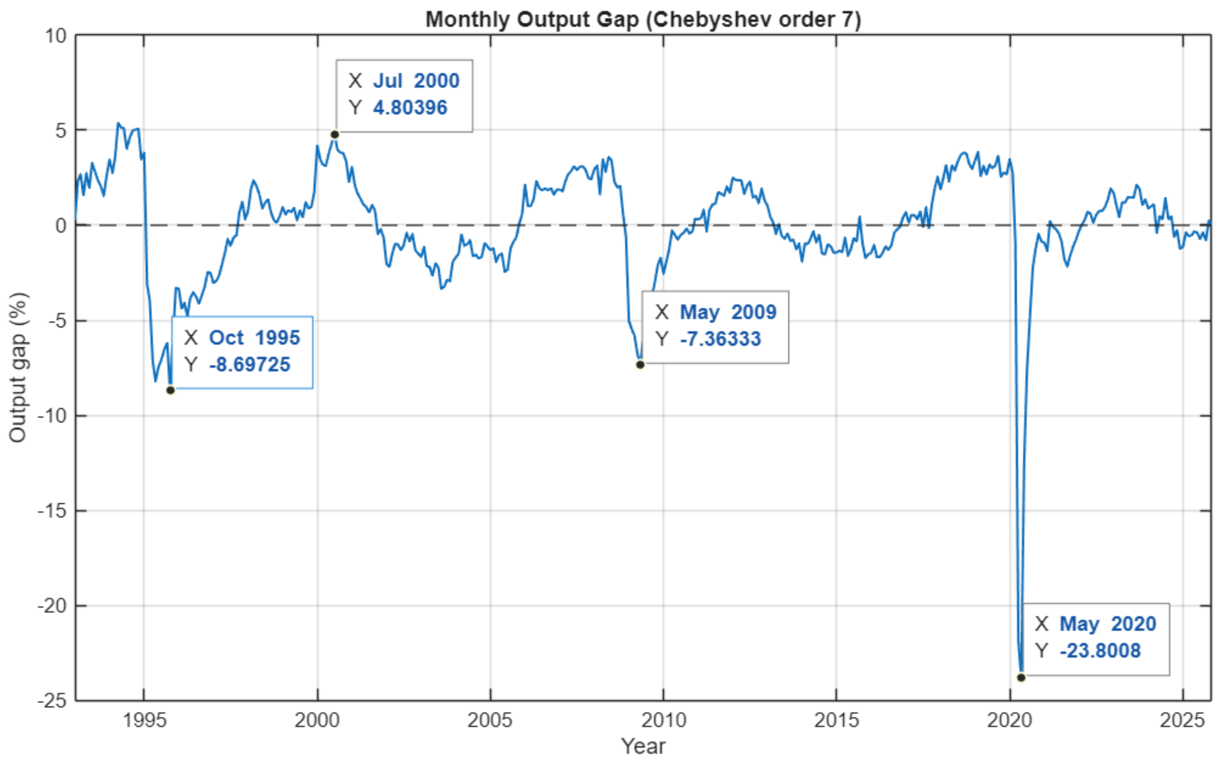

Figure 1: Monthly output gap for Mexico using A 7th degree Chebyshev polynomial(1993-2025)

The series exhibits well-defined cyclical fluctuations around zero, capturing periods of economic expansion and contraction. It is interesting to note that large negative gaps are observed during economic crisis, such as the crisis of 2008-2009 and the COVID-19 pandemic in 2020.

Moreover, Figure 2 shows the trend estimated by the model together with the observed values of $\log(\text{IGAE}_t)$.

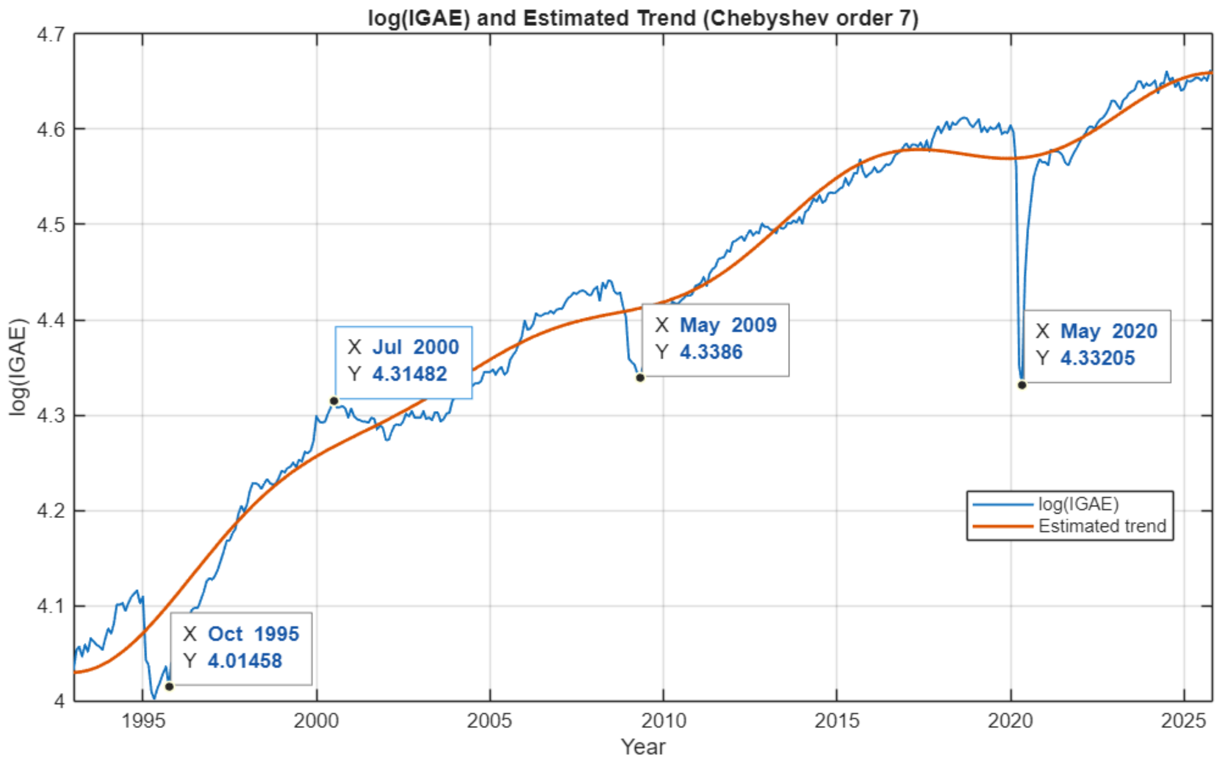

Figure 2: Estimated trend using 7th-degree Chebyshev polynomial (Mexico, 1993-2025)

(2) Comparison with Banco de México's estimates.

Using the data provided by Banco de México, the results obtained from the previously estimated model can be compared with the official output gap series.

It is important to note that Banco de México reports the output gap at a quarterly frequency, whereas the model-based estimates in the previous section are monthly. Therefore, only those observations from the Chebyshev-based model that coincide with the quarterly measurement dates were retained for the comparison. In addition, Banco de México's series is available only up to 2021, so the sample of the model-based estimates was truncated accordingly.

Thus, Figure 3 displays the output gap estimated using a degree-7 Chebyshev polynomial together with the output gap series published by Banco de México.

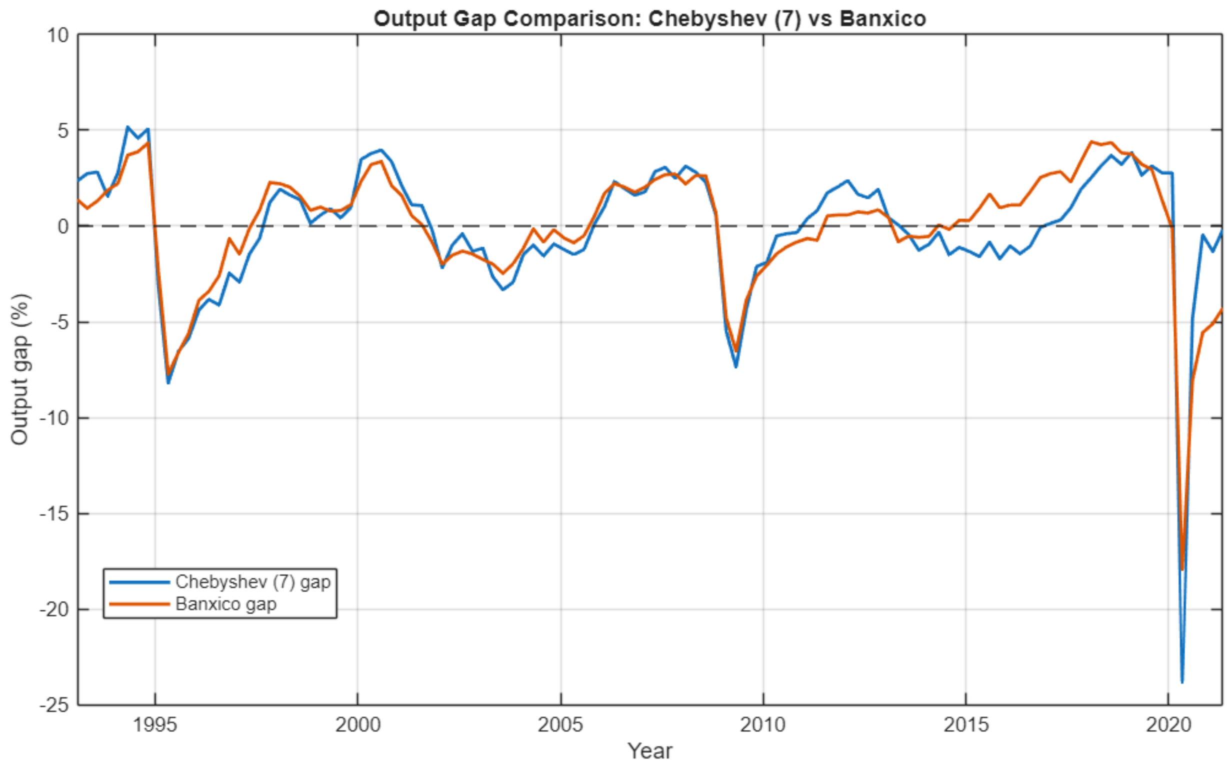

Figure 3: Output Gap comparison between the 7th degree polynomial and the data from BANXICO (1993-2021)

Overall, both series exhibit a high degree of comovement, capturing the main fluctuations in Mexico. In particular, both measures identify pronounced negative gaps during major economic downturns, such as the mid-1990s crisis, the global financial crisis of 2008–2009, and the COVID-19 pandemic in 2020. While some differences are observed, specially around 2015, the similarity in overall dynamics suggests that the Chebyshev-based approach provides a reasonable approximation to the official output gap measure.

Additionally, a scatterplot was made to examine the linear association between the two output gap measures (Figure 4). The plot shows a strong positive relationship between both estimates. In fact, the correlation coefficient between the two data sets is 0.9051, indicating a high degree of linear comovement.

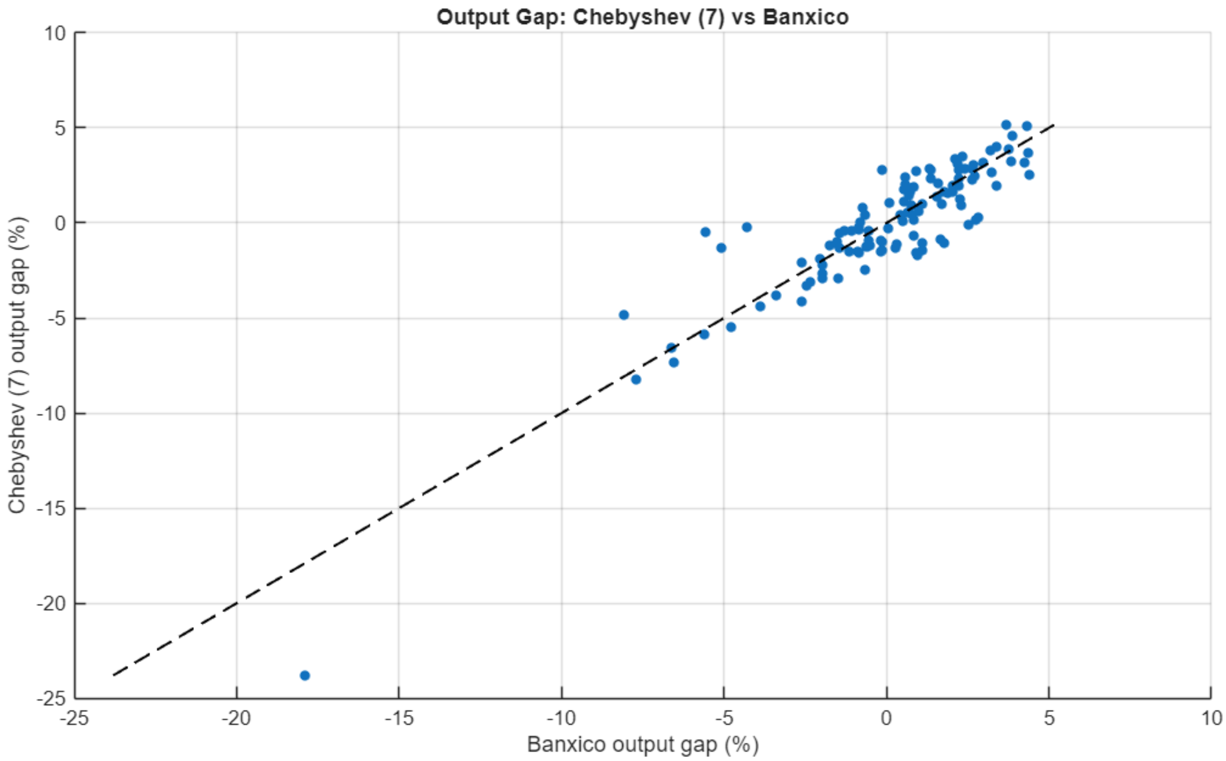

Figure 4: Scatterplot between the 7th degree Chebyshev polynomial and the data from BANXICO

(3) Output gap for Mexico using a Chebyshev polynomial of order 51.

Finally, a Chebyshev polynomial of order 51 was also considered. Figure 5 presents the resulting estimates of the monthly output gap for Mexico (1993–2025).

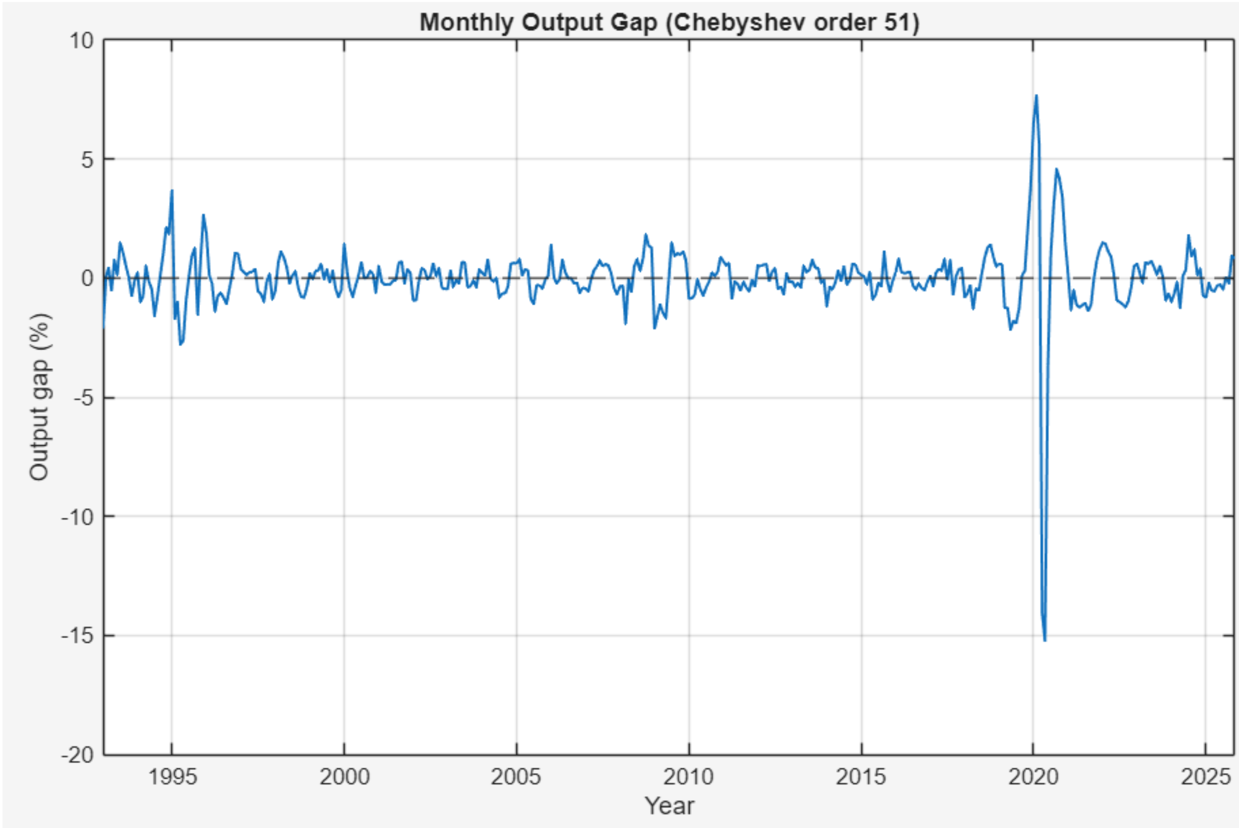

Figure 5: Monthly output gap for Mexico using A 51th degree Chebyshev polynomial(1993-2025)

Observe that the monthly output gap is very close to zero in the most part. This is because using a Chebyshev polynomial of order 51 gives the model a great amount of freedom and, therefore, the estimated trend is no longer a smooth curve, but rather a function that follows the data very closely (Figure 6).

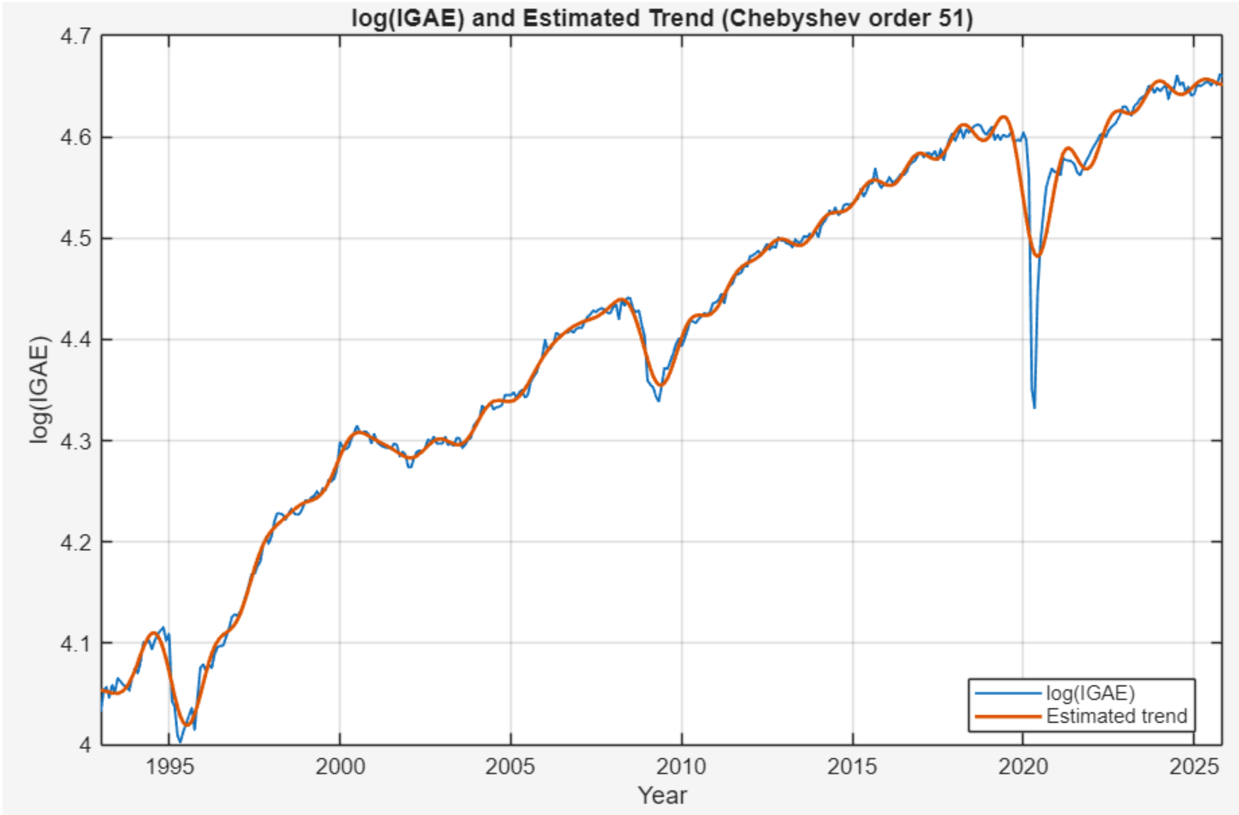

Figure 6: Estimated trend using 51th-degree Chebyshev polynomial (Mexico, 1993-2025)

Because the trend adapts so tightly to the observed series, many short-term movements that would normally be interpreted as cyclical fluctuations are absorbed into the trend component. As a consequence, the remaining output gap becomes relatively small and oscillates around zero for most of the sample.

Although the order-51 polynomial fits the data extremely well, it tends to produce an output gap that is less informative about the short-term cycles. In comparison, lower-order polynomials impose more smoothness on the trend and, therefore, capture better the short-term fluctuations. In fact, when the output gap estimated using the order-51 Chebyshev polynomial is compared with the official data provided by Banco de México, the results are noticeably less similar. Unlike the high correlation obtained with the order 7 model, the correlation in this case is only 0.4297, which is significantly lower.